



Cairo University
 Faculty of Engineering
 Credit Hours System
Communications and Computer Engineering Department

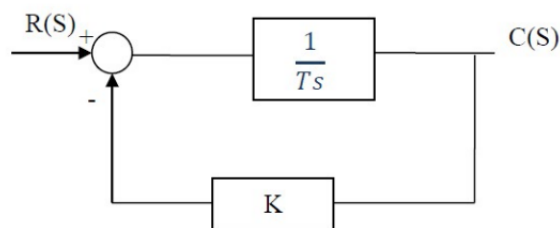
Fall 2026

Course Code EECS 304
 Course Title Control I
 Course Instructors Dr. Ragia Badr - Dr. Nasr Antar
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Sheet 3: Steady State Analysis

Question (1):

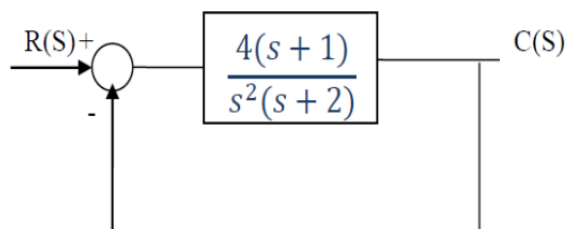
For the following system, find:



1. The transfer function $C(s)/R(s)$.
2. The output $c(t)$ if:
 - (a) $R(s)$ is step input.
 - (b) $R(s)$ is ramp input.
 - (c) $R(s)$ is parabolic input.
3. The steady state output c_{ss} and the steady state error e_{ss} , if possible.

Question (2):

For the following system, find:



1. The position, velocity and acceleration error constants.
2. The steady state error when the input is

$$R(s) = \frac{3}{s} - \frac{1}{s^2} + \frac{1}{2s^3}$$

Question (3):

Find the transfer function for the following systems, then get the steady state value of the output for input $r(t) = 1 + 2t$:

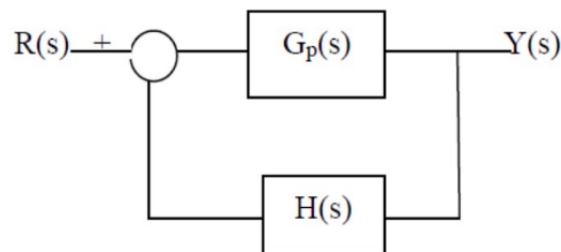
1. $\ddot{y}(t) + 3\ddot{y}(t) + \dot{y}(t) - 2y(t) = r(t)$
2. $y^{(4)} + 2\ddot{y}(t) + 5\ddot{y}(t) + \dot{y}(t) + 2y(t) = r(t)$

Question (4):

A feedback system has the following transfer functions:

$$G_p(s) = \frac{K}{(Ts + 1)}, \quad H(s) = \frac{1}{s}$$

Determine the static error coefficients for this system.

**Question (5):**

A unity feedback system has a forward transfer function of:

$$G_p(s) = \frac{12(s + 4)}{s(s + 1)(s + 3)(s^2 + 2s + 10)}$$

1. Determine the static error coefficients for this system.
2. Determine the steady state error and the steady state output for a reference input $r(t) = 16 + 2t$ and for $r(t) = 5t^2$